

Section 6.1 Exercises

Directions: For exercises 1 - 9, graph one full period of each function, starting at $x = 0$. State the amplitude, period, and midline. State a maximum and minimum y -value and their corresponding x -values on this period.

1. $f(x) = 2\sin x$

2. $f(x) = \frac{2}{3}\cos x$

3. $f(x) = -3\sin x$

4. $f(x) = 4\sin x$

5. $f(x) = 2\cos x$

6. $f(x) = \cos(2x)$

7. $f(x) = 2\sin\left(\frac{1}{2}x\right)$

8. $f(x) = 4\cos(\pi x)$

9. $f(x) = 3\cos\left(\frac{6}{5}x\right)$

Directions: For exercises 10 - 14, graph one full period of each function. For each function, state the amplitude, period, and midline. State a maximum and minimum y -value and their corresponding x -values on this period. State the phase shift and vertical translation, if applicable.

10. $y = 2\sin\left(x - \frac{5\pi}{6}\right)$

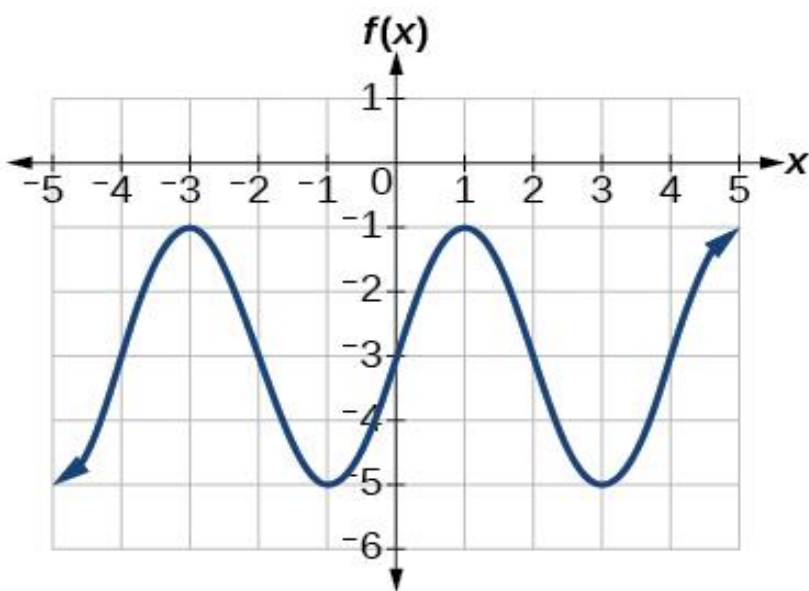
11. $y = -\cos\left(x + \frac{\pi}{3}\right) + 1$

12. $y = 4\cos\left(2\left(x + \frac{\pi}{4}\right)\right) - 3$

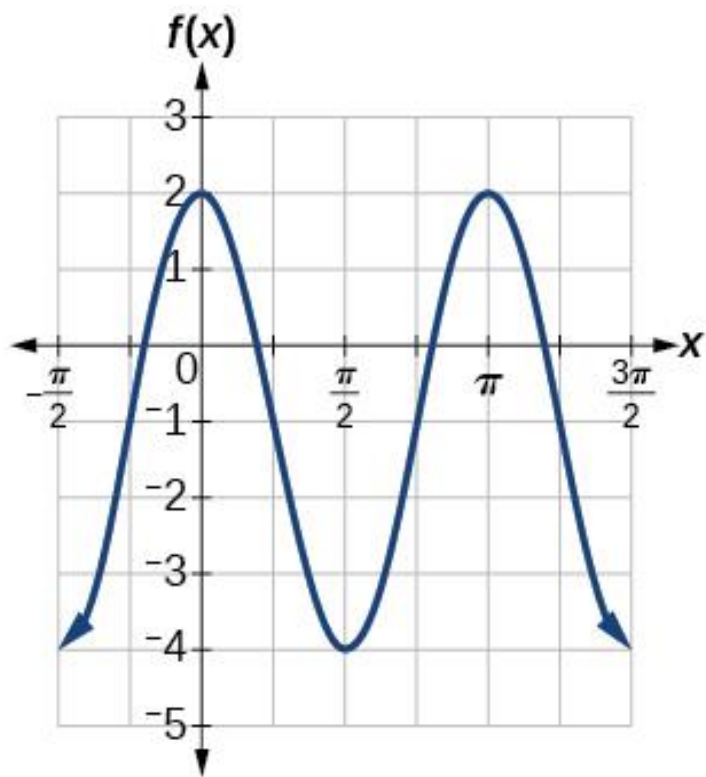
13. $y = -\sin\left(\frac{1}{2}x + \frac{5\pi}{3}\right)$

14. $y = 4\sin\left(\frac{\pi}{2}(x - 3)\right) + 7$

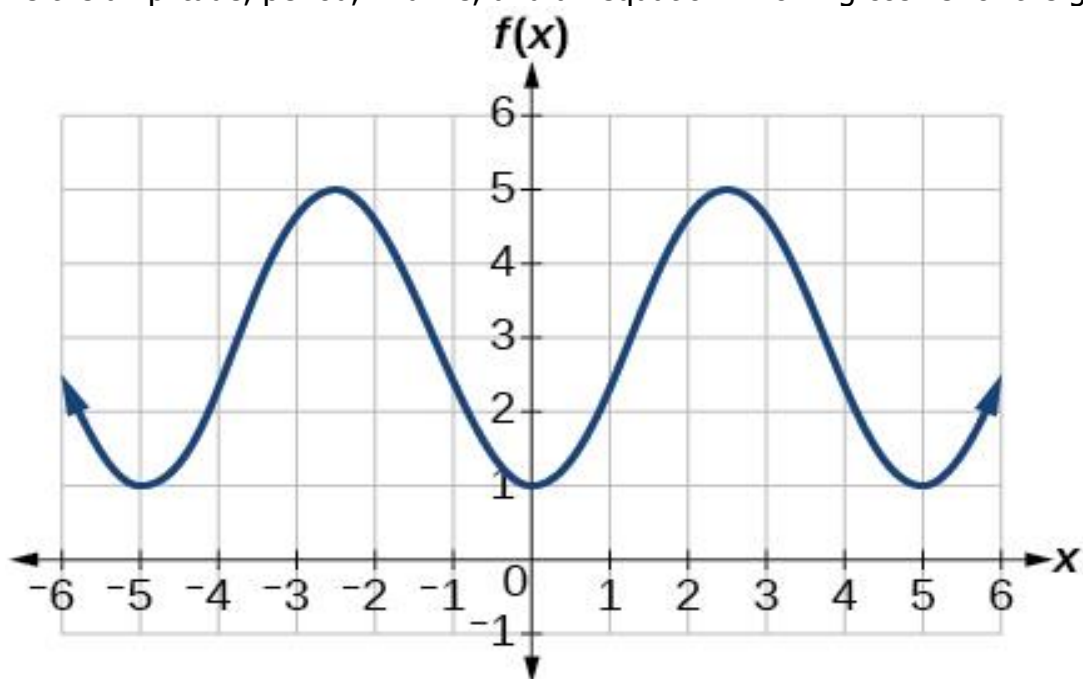
15. Determine the amplitude, midline, period, and an equation involving the sine for the graph below.



16. Determine the amplitude, period, midline, and an equation involving cosine for the graph below.



17. Determine the amplitude, period, midline, and an equation involving cosine for the graph below.



18. Determine the amplitude, period, midline, and an equation involving sine for the graph below.

